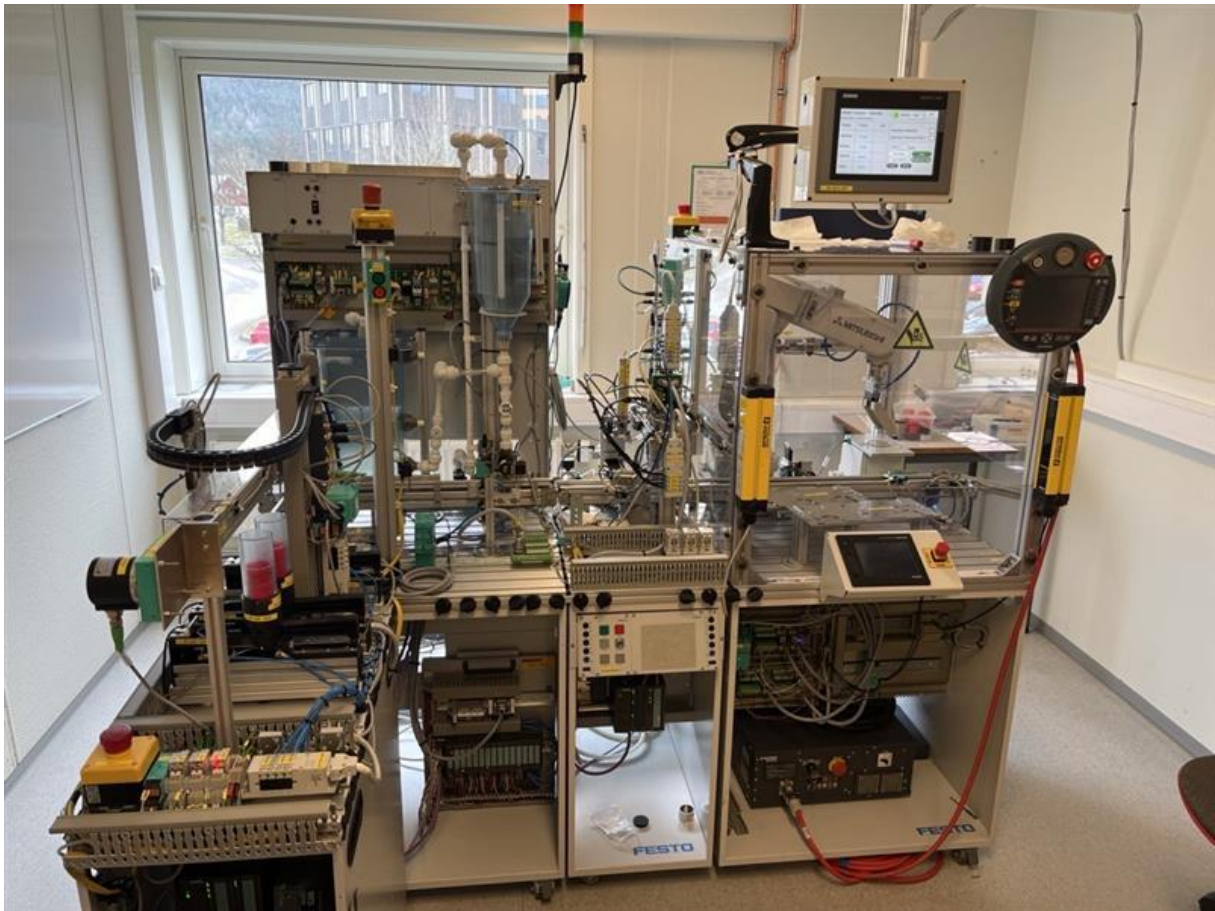




Figur 1 Festo anlegge

Opprett ordre

Rød (antall)

Svart (antall)

Testfase



Before = før optimalisering, After = etter optimalisering

Total: **10/12**

Vann % per cup

Bare cup 1..Total vises. Resten skjules automatisk.

Red 1

Red 2

Red 3

Red 4

Red 5

Black 1

Black 2

Black 3

Black 4

Black 5

Lagre ordre

Regel: Rød 0–6, Svart 0–6, Total 0–12.

Figur 2 Skjerm bilde av ERP-systemet, der operatøren kan opprette en ordre.

```
C:\Users\maxam\OneDrive\Di X + v

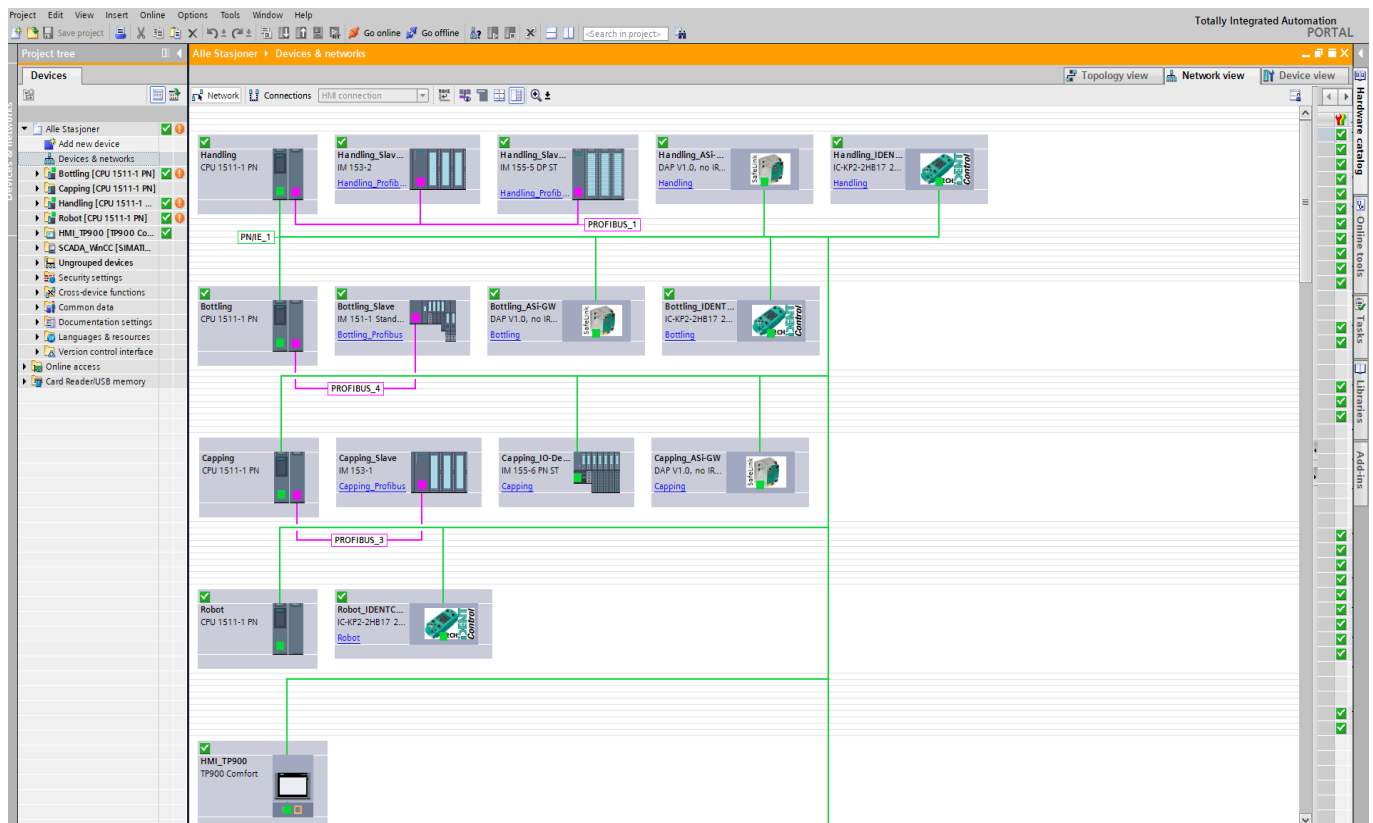
=== ERP Gateway ===
Handling PLC: 192.168.0.201
Robot PLC: 192.168.0.211
[11:10:45] Connecting PLC 192.168.0.211...
[11:10:45] PLC connected: 192.168.0.211
[11:10:45] OrderDone TRUE but no Running order found.
[11:13:15] Sending OrderId=22 to PLC... waiting for OrderAck=TRUE
[11:13:15] PLC ack received. SQL: Order 22 -> Ready
[11:16:30] SQL: Order 22 -> Running + StartedDate set
[11:16:30] Orders updated -> OrderId=22, QuantityProduced=0
[11:16:30] Orders updated -> OrderId=22, QuantityProductFault=0
[11:17:09] Orders updated -> OrderId=22, QuantityProductFault=1
[11:17:09] Robot QC saved -> OrderId=22, JarId=1, WeightDifference=-1,424, RealWeight=14,587, TimeSpentMinutes=0,02
[11:17:21] Orders updated -> OrderId=22, QuantityProductFault=2
[11:17:21] Robot QC saved -> OrderId=22, JarId=2, WeightDifference=-1,801, RealWeight=15,611, TimeSpentMinutes=0,1
[11:17:28] Orders updated -> OrderId=22, QuantityProductFault=3
[11:17:28] Robot QC saved -> OrderId=22, JarId=3, WeightDifference=-1,682, RealWeight=17,131, TimeSpentMinutes=0,12
[11:17:35] Orders updated -> OrderId=22, QuantityProductFault=4
[11:17:36] Robot QC saved -> OrderId=22, JarId=4, WeightDifference=-2,082, RealWeight=18,132, TimeSpentMinutes=0,12
[11:17:43] Orders updated -> OrderId=22, QuantityProductFault=5
[11:17:43] Robot QC saved -> OrderId=22, JarId=5, WeightDifference=-2,423, RealWeight=19,192, TimeSpentMinutes=0,1
[11:18:13] Orders updated -> OrderId=22, QuantityProductFault=6
[11:18:14] Robot QC saved -> OrderId=22, JarId=6, WeightDifference=-2,236, RealWeight=20,78, TimeSpentMinutes=0,5
[11:18:21] Orders updated -> OrderId=22, QuantityProductFault=7
[11:18:22] Robot QC saved -> OrderId=22, JarId=7, WeightDifference=-2,865, RealWeight=21,552, TimeSpentMinutes=0,5
[11:18:29] Orders updated -> OrderId=22, QuantityProduced=1
[11:18:30] Robot QC saved -> OrderId=22, JarId=8, WeightDifference=-0,812, RealWeight=25,006, TimeSpentMinutes=0,5
[11:18:38] Orders updated -> OrderId=22, QuantityProduced=2
[11:18:38] Robot QC saved -> OrderId=22, JarId=9, WeightDifference=-0,74, RealWeight=26,479, TimeSpentMinutes=0,5
[11:18:46] Order 22 marked for finish. Waiting 2 seconds for final PLC updates...
[11:18:46] Orders updated -> OrderId=22, QuantityProduced=3
[11:18:47] Robot QC saved -> OrderId=22, JarId=10, WeightDifference=0,415, RealWeight=29,035, TimeSpentMinutes=0,5
[11:18:48] SQL: Order 22 -> Finished + FinishedDate + TimeSpentMinutes (using actual OrderDone time)
```

Figur 3 Skjerm bilde av gateway-programmet.

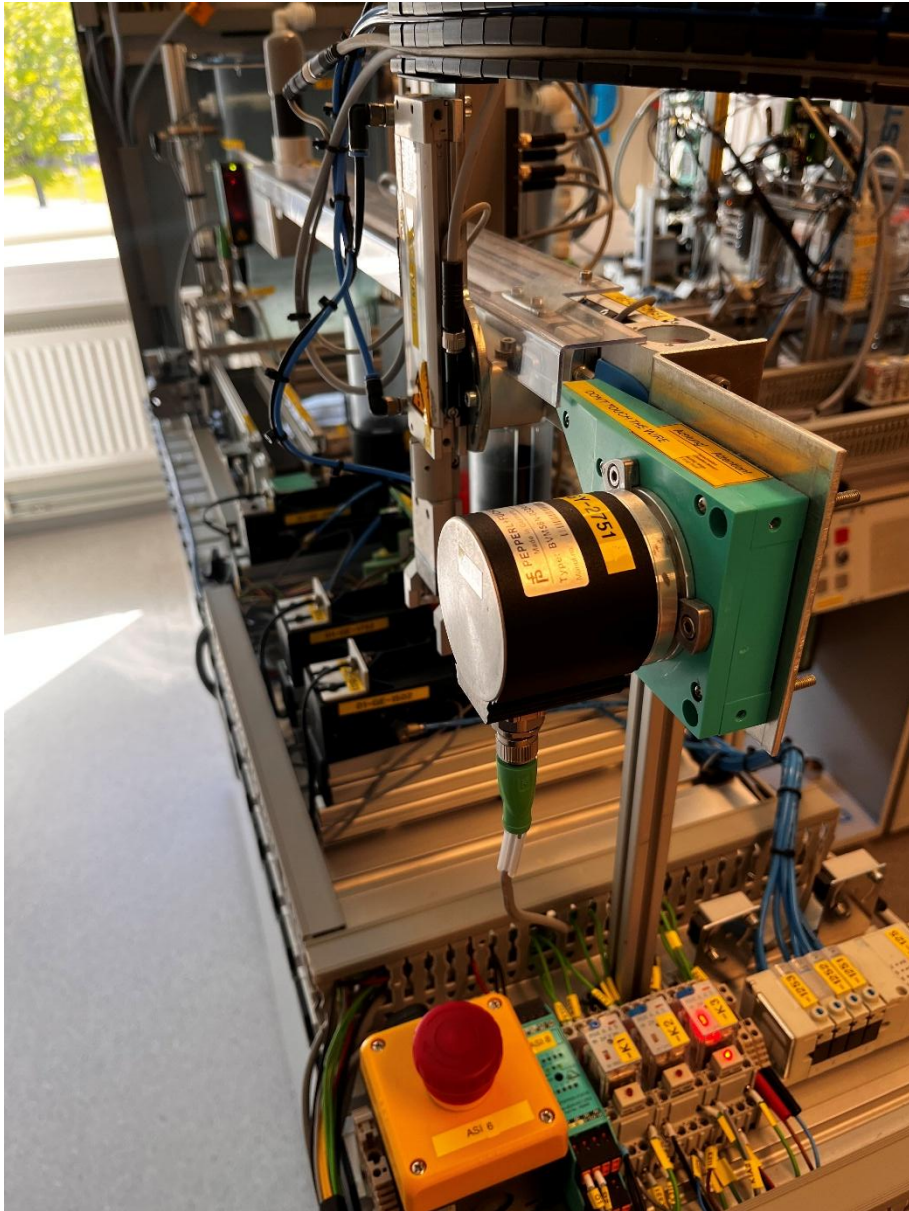
20	2	5	5	10	Finished	2026-04-14 11:02:40.790	2026-04-14 11:03:18.873	2026-04-14 11:05:44.000	10	0	2.42	Before
21	1	5	5	10	Finished	2026-04-14 10:53:42.143	2026-04-14 10:54:06.797	2026-04-14 10:57:07.460	10	0	3.01	Before

	CupId	OrderId	CupNo	Color	WaterPercent	WeightDifference	RealWeight	TimeSpentMinutes
192	12	2	2	Red	20	0.1797514	17.83225	0.72
193	13	2	3	Red	30	0.1750813	19.23792	0.70
194	14	2	4	Red	40	0.1123266	20.70167	0.68
195	15	2	5	Red	50	0.003105164	22.21189	0.70
196	16	2	6	Black	60	-0.1959257	23.42007	0.73
197	17	2	7	Black	70	-0.0170002	25	0.72
198	18	2	8	Black	80	-0.1052675	26.31273	0.70
199	19	2	9	Black	90	0.5383415	28.35734	0.72
200	20	2	10	Black	100	0.2409658	29.46097	0.73

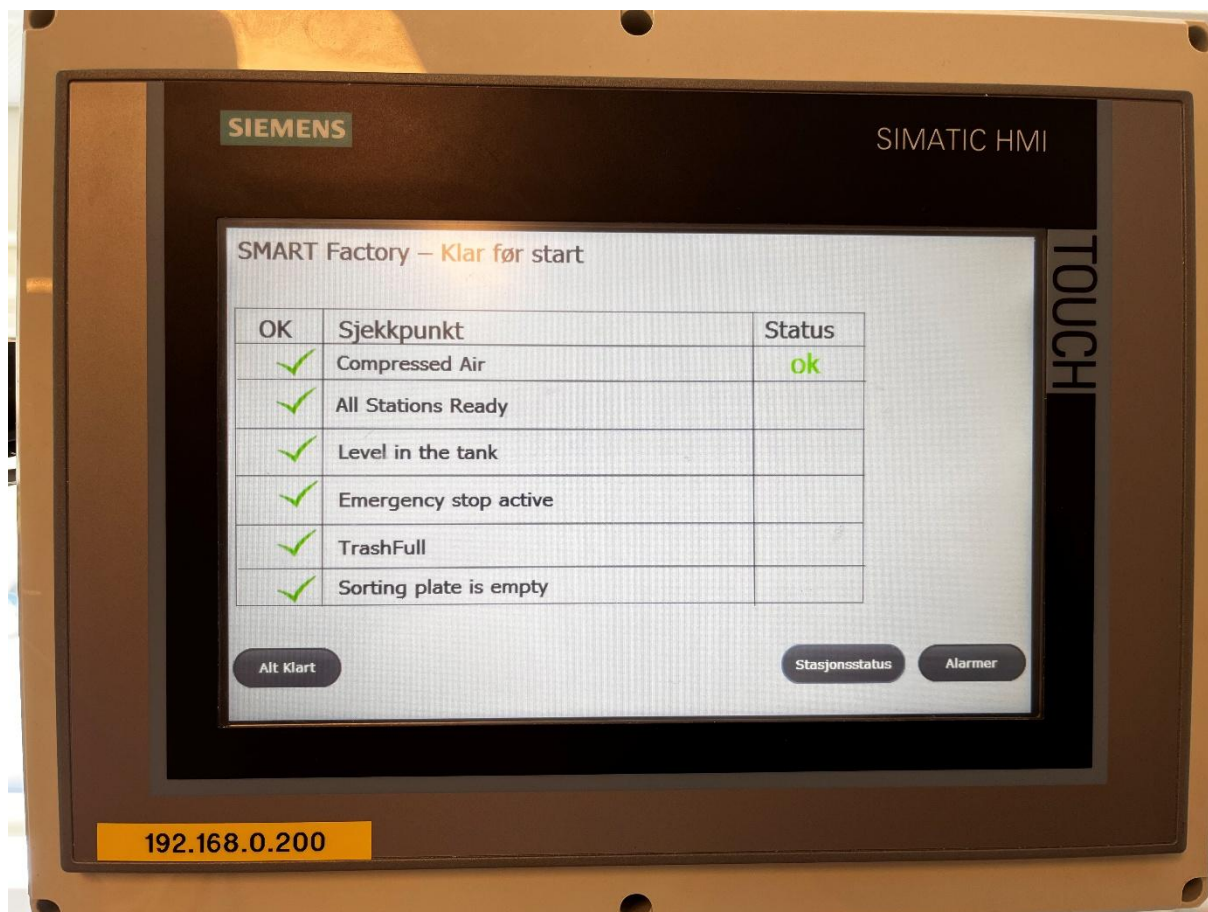
Figur 4 Skjerm bilde av SQL-databasen.



Figur 5 Skjerm bilde fra TIA Portal V17 med oversikt over alle komponenter.



Figur 6 Bilde av enkoder på handling-stasjonen.



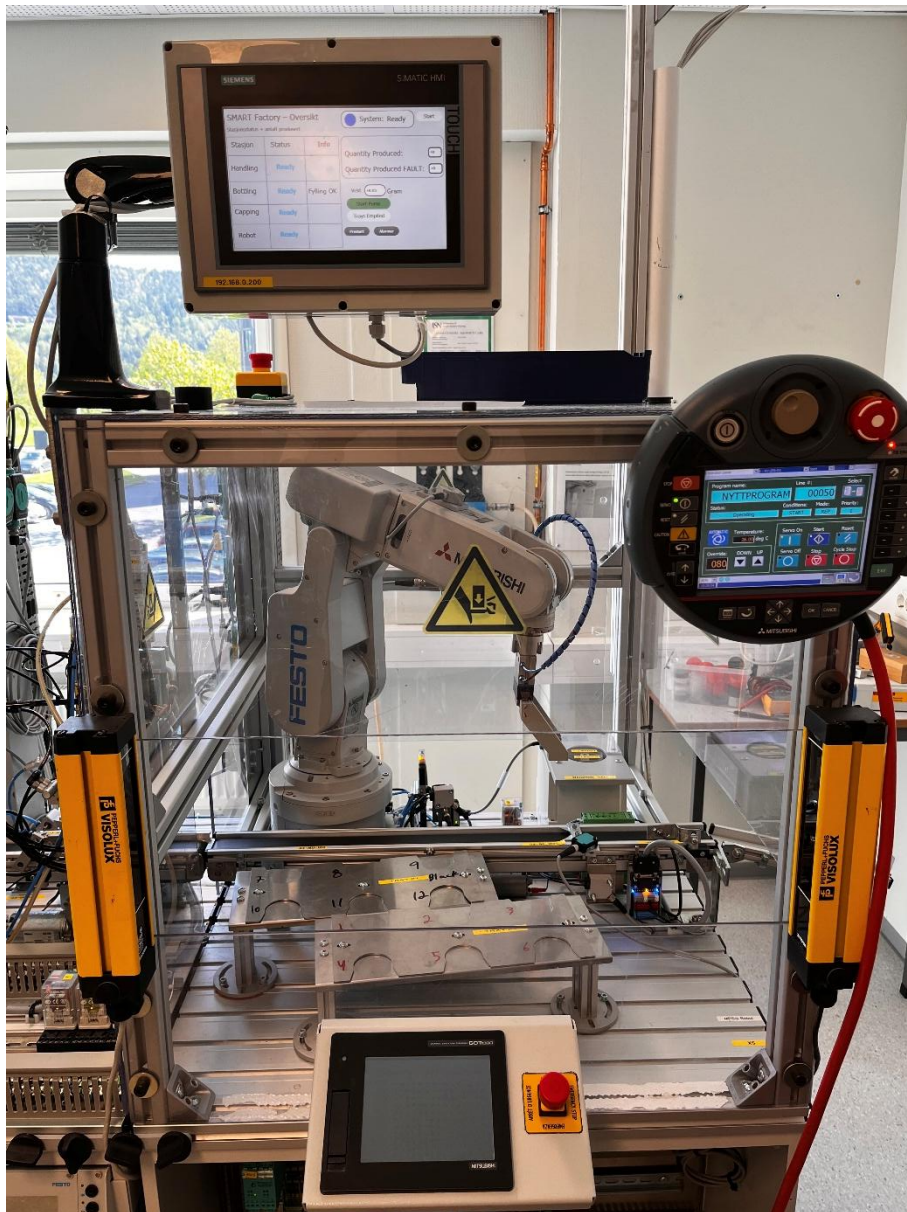
Figur 7 HMI som viser at anlegget er klart for start.



Figur 8 Robotens fjernkontroll.



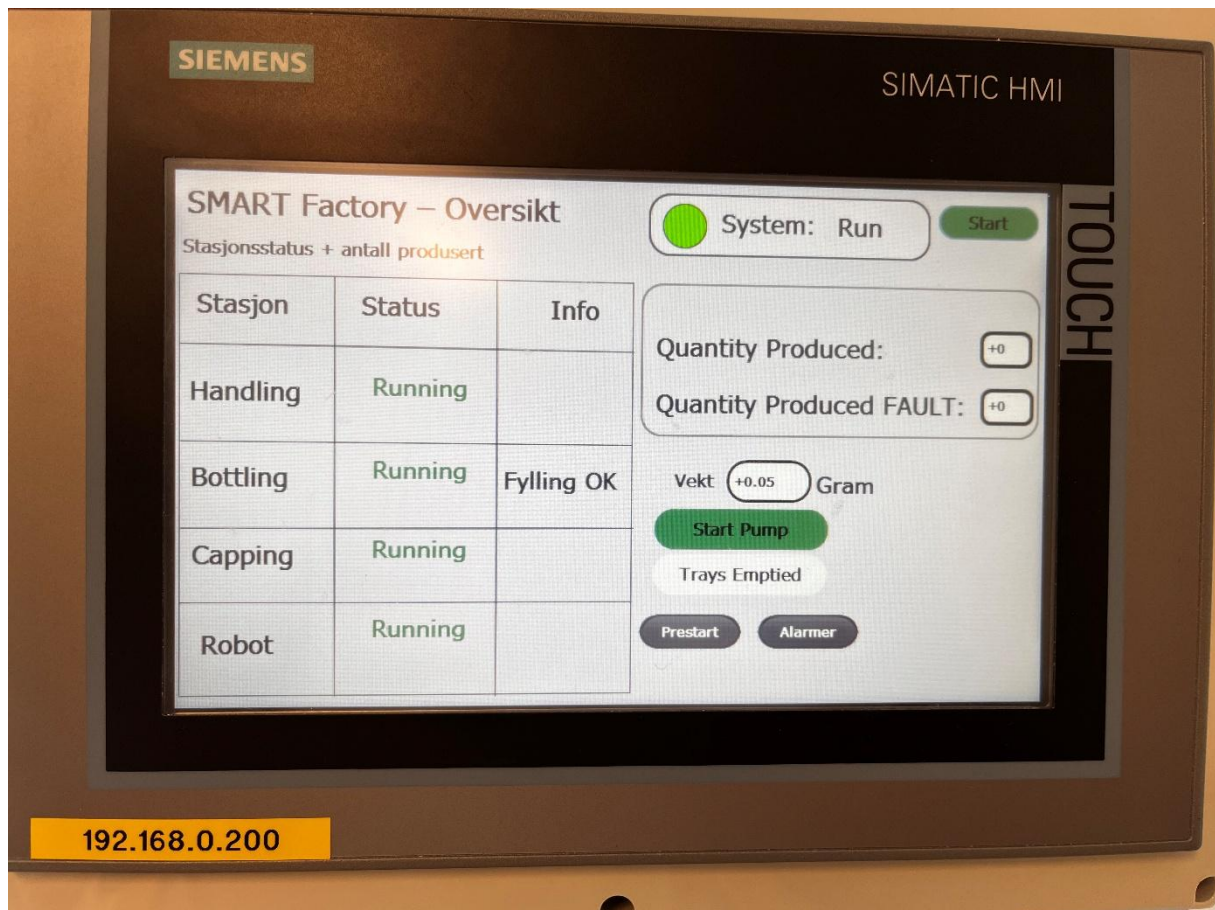
Figur 9 Mitsubishi robotkontroller CR750-02-VD1-1-S15.



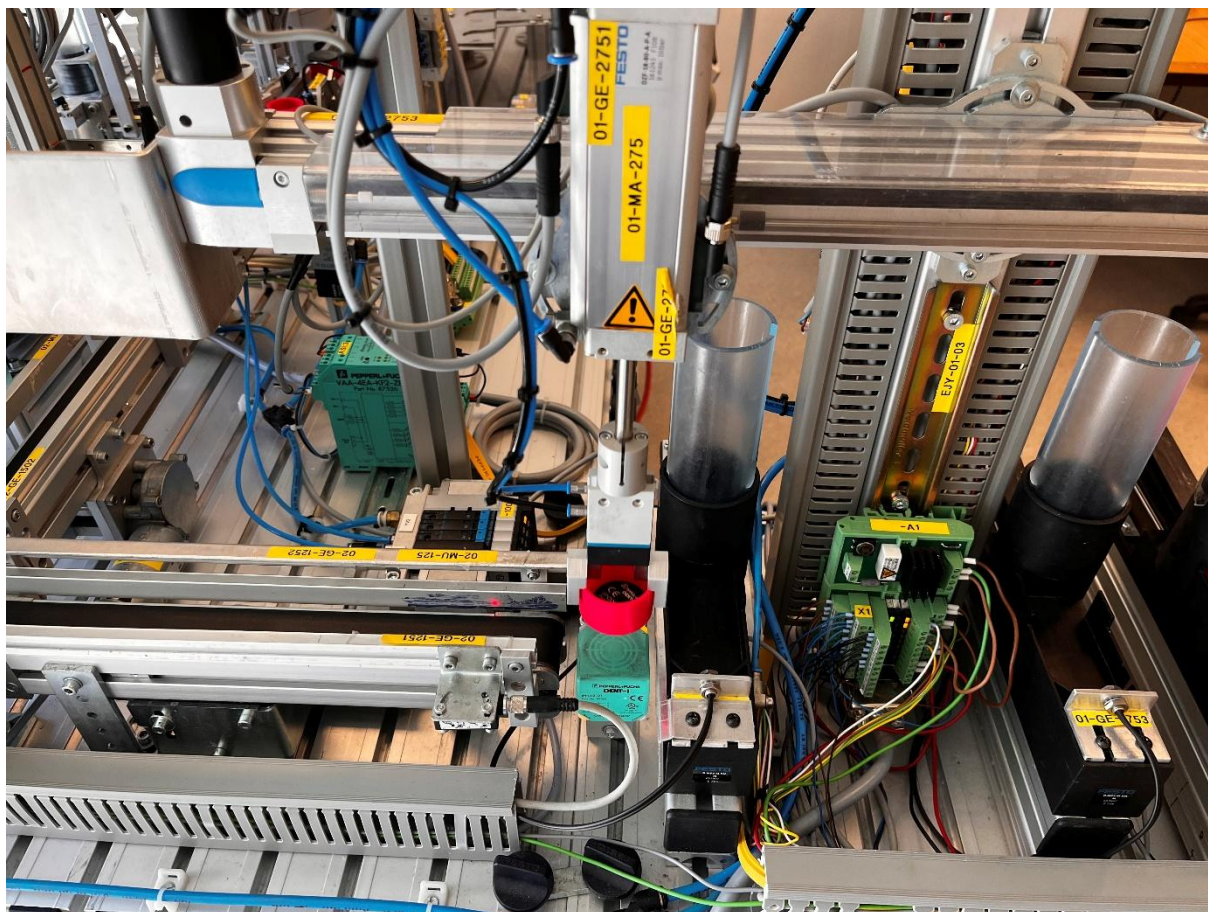
Figur 10 Robot Stasjon.



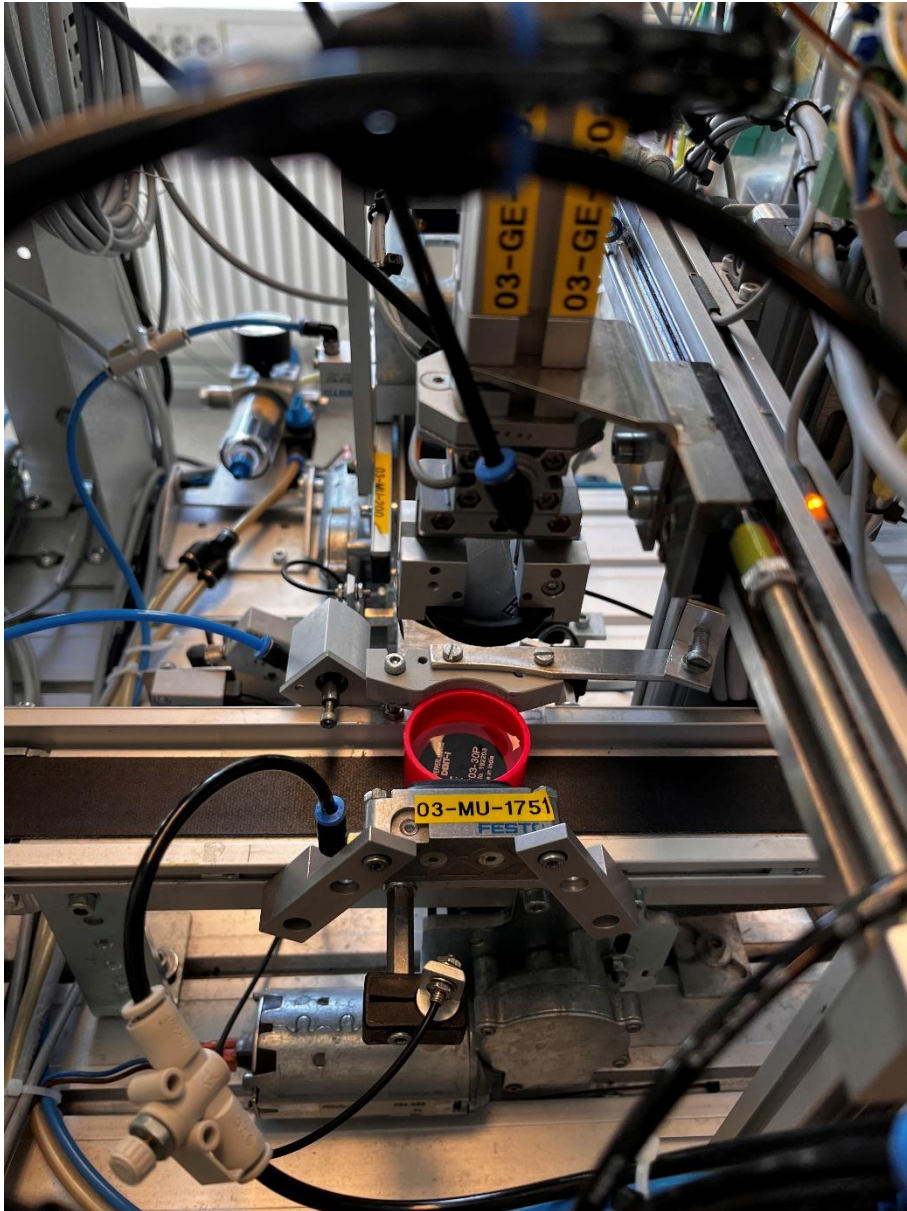
Figur 11 HMI-oversikt



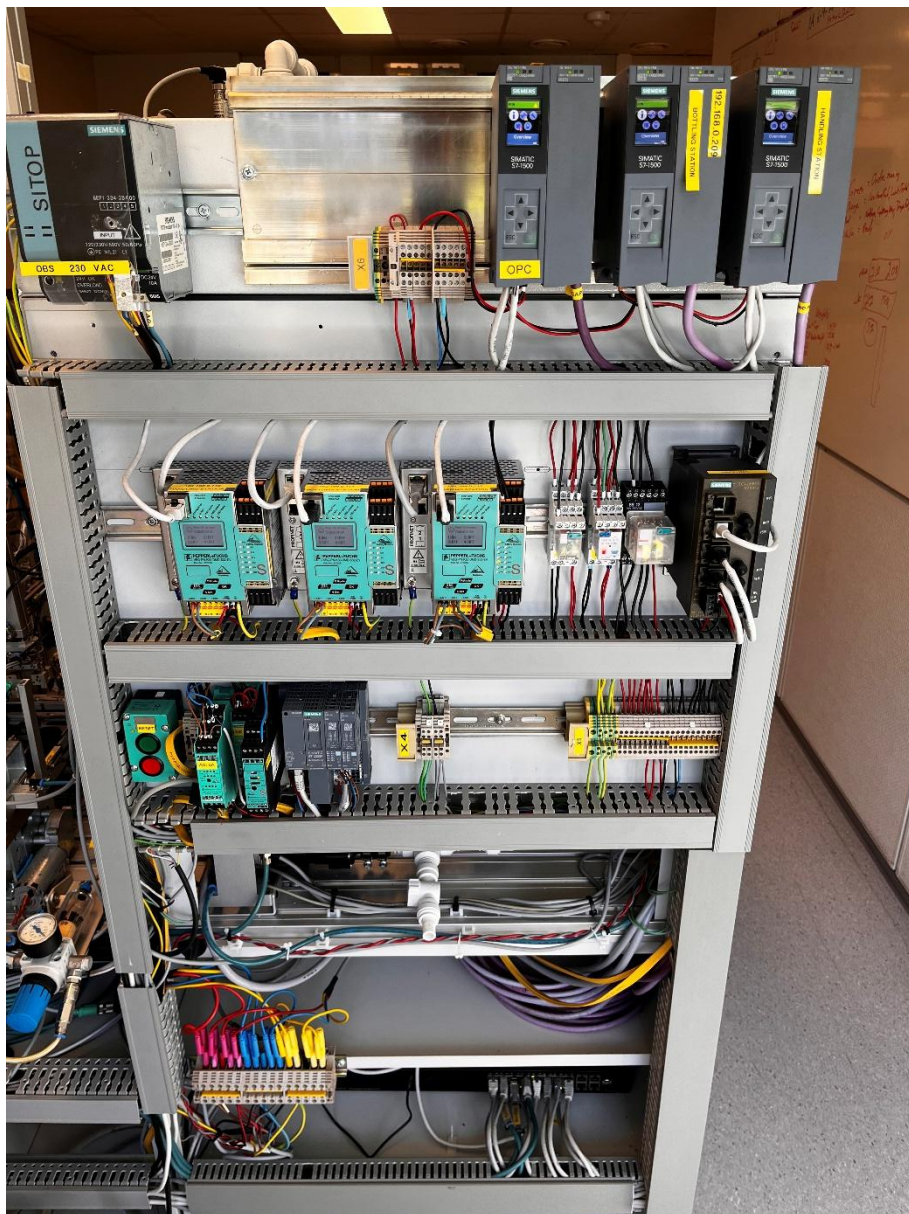
Figur 12 HMI som viser Festo-anlegget i drift.



Figur 13 Kopper ved RFID-leser på handling-stasjonen.



Figur 14 Kopp etter lukking.



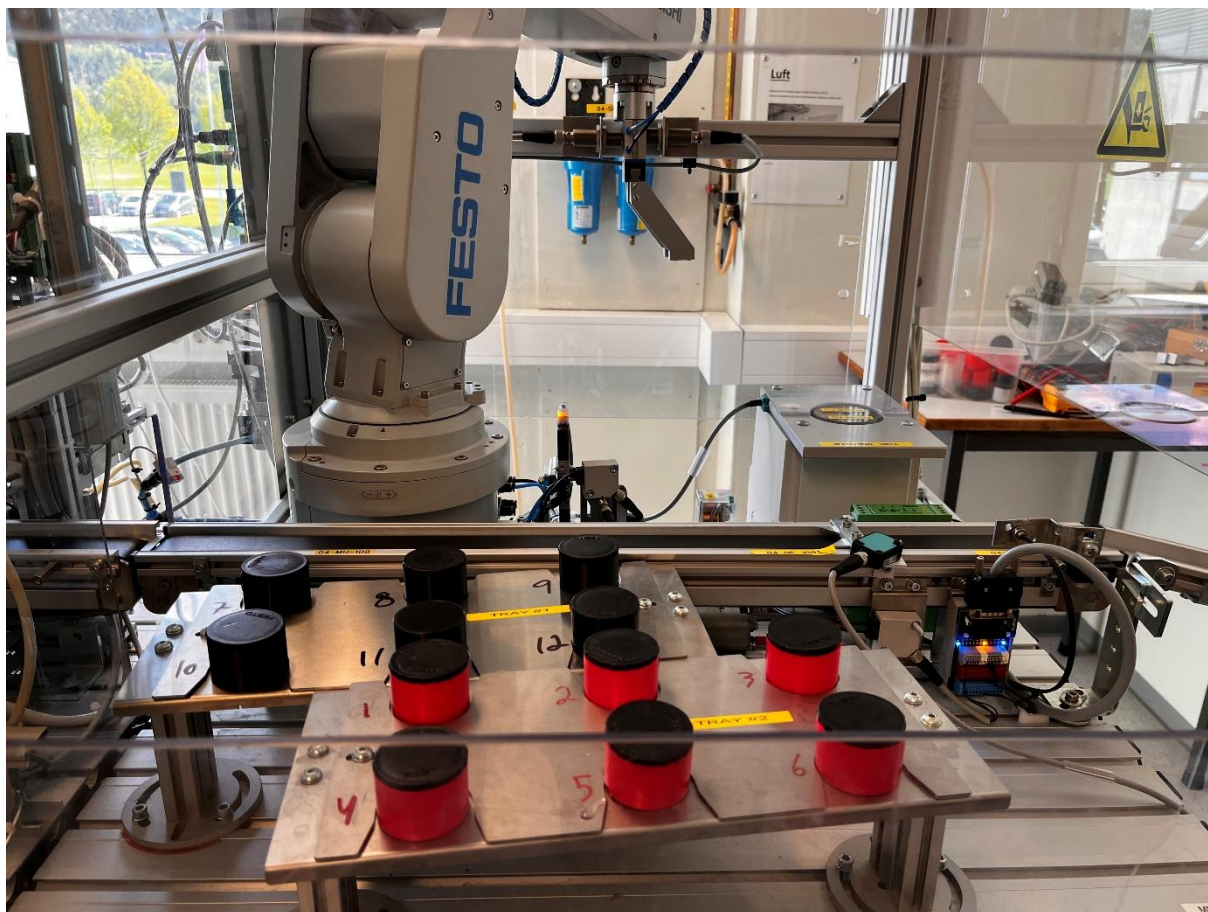
Figur 15 Bilde av PLS S7-1500, ASI Gateway, Safe link og Nettverk Switch helt ned



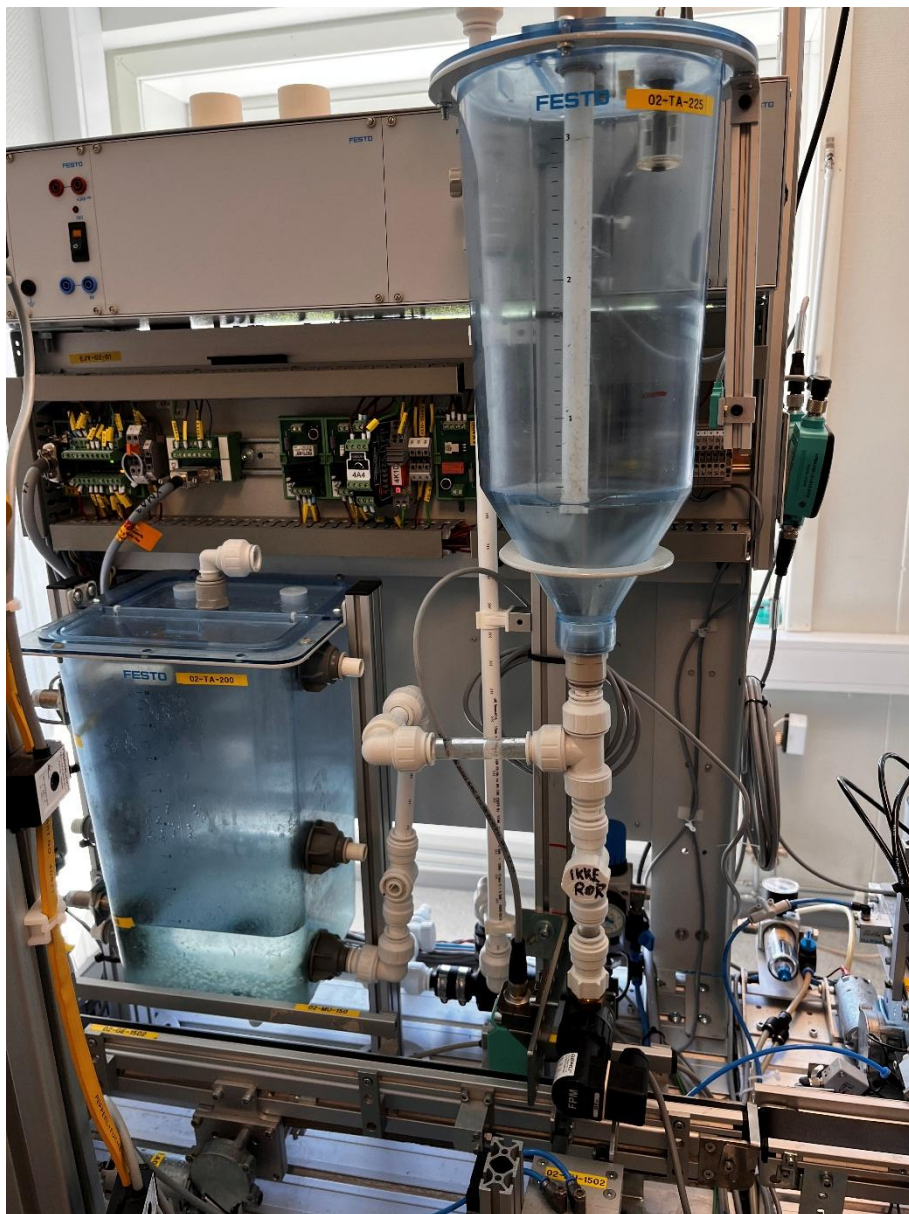
Figur 16 Nødstopp på bottling-stasjonen med reset-knapp



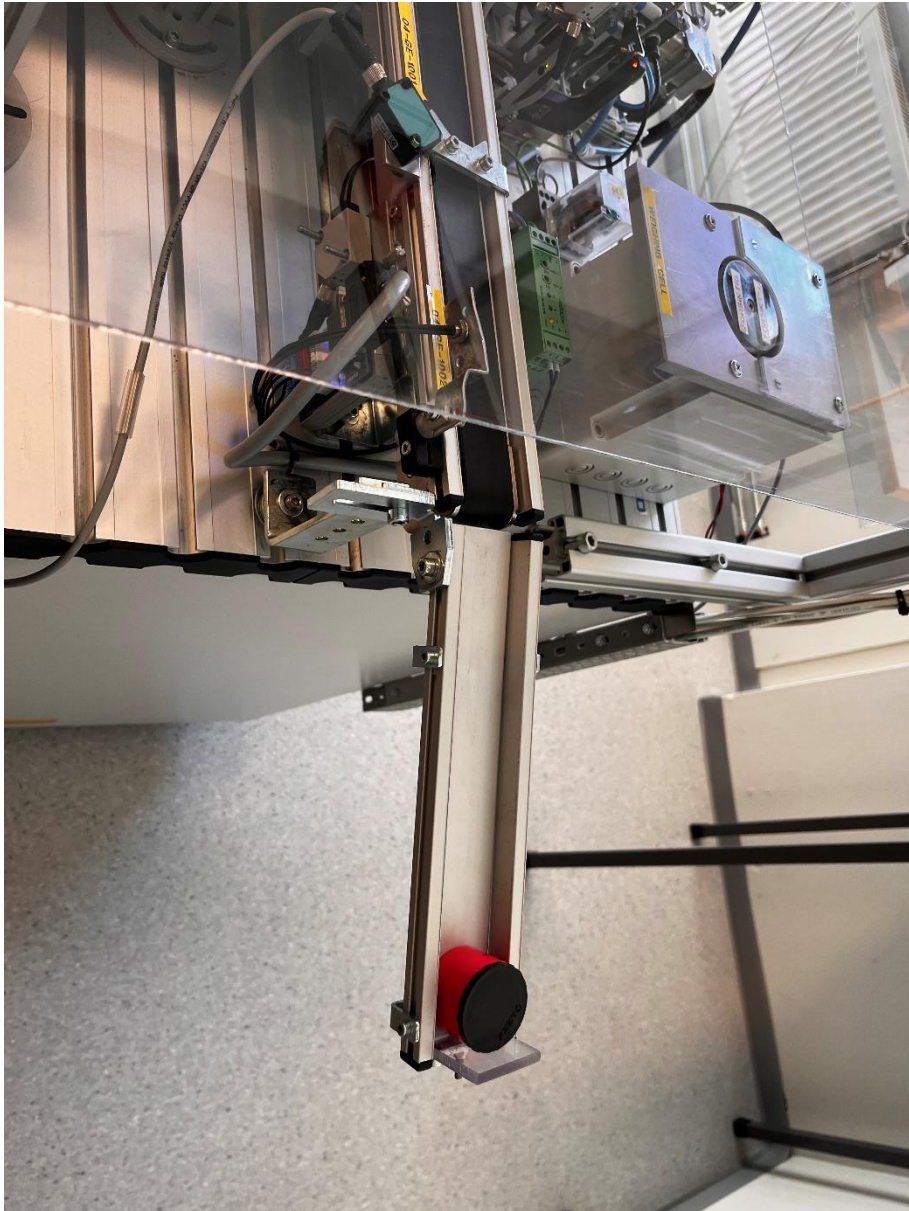
Figur 17 Robot i sikker posisjon. Operatøren kan da føre hånden inn i robotstasjonen uten at lysgardinen utløser nødstop.



Figur 18 : Ferdig ordre med roboten i sikker posisjon.



Figur 19 Tank til fylling.



Figur 20 Kopper som ikke er godkjent.



Figur 21 Luftkompressor.



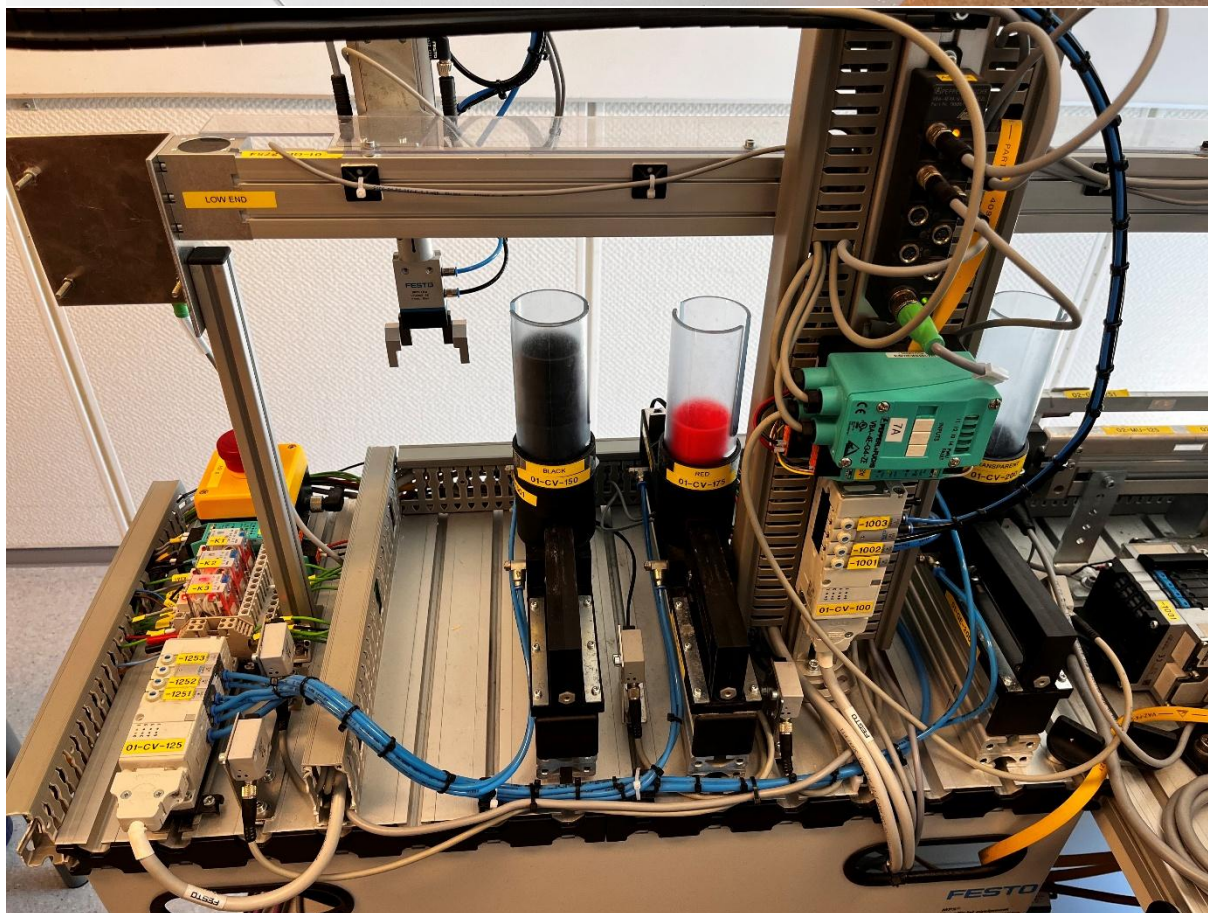
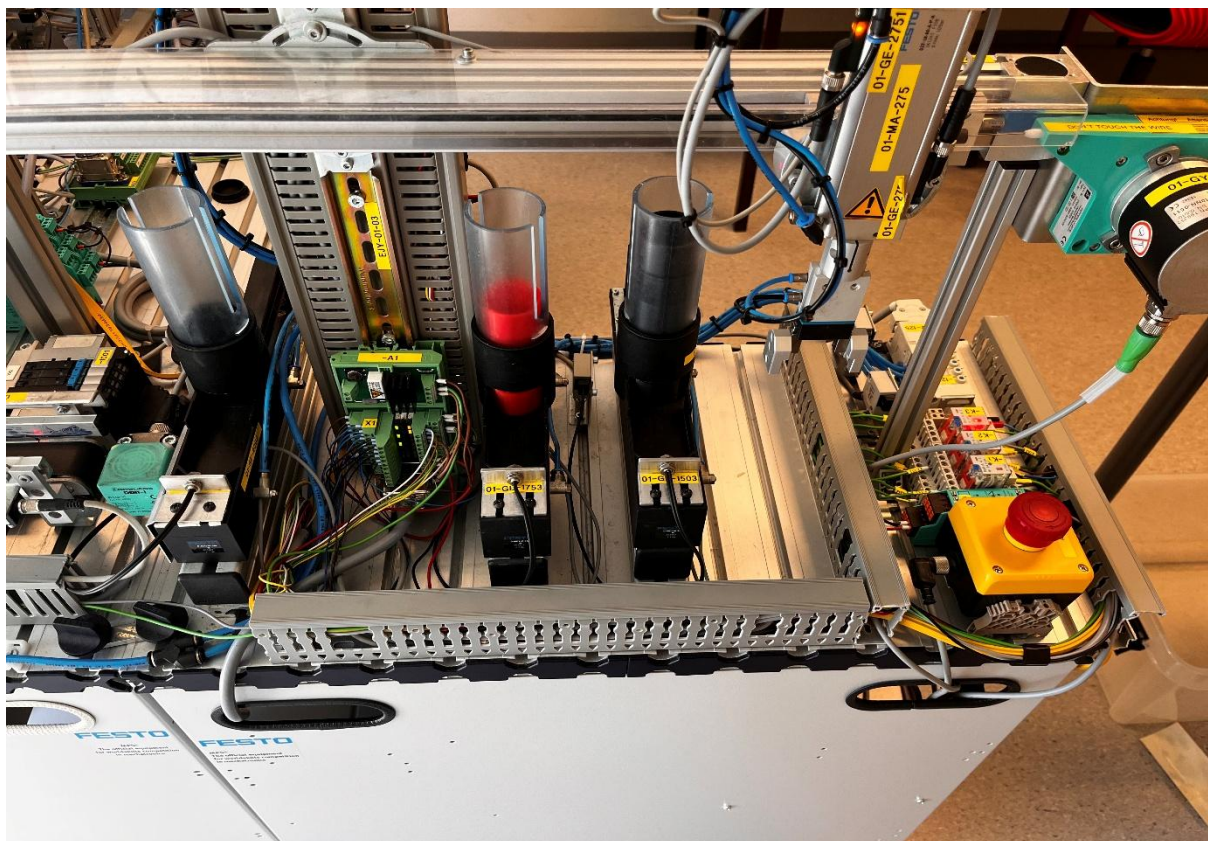
Figur 22 Vekt/veiecelle for veiing, RFID-lesing og i robotstasjonen.



Figur 23 Robot stasjon fra bak



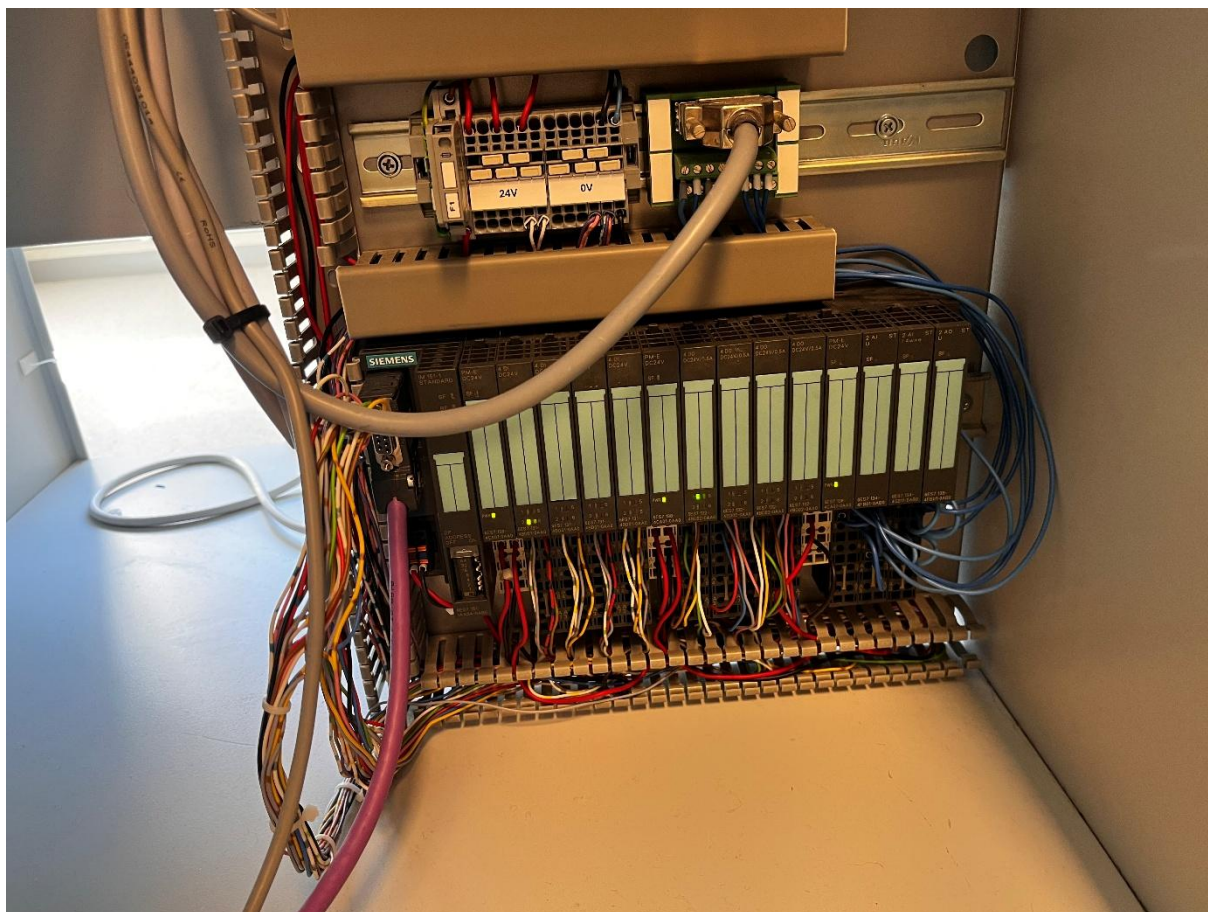
Figur 24 Robot-PLS S7-1500 for tilbakemelding og robotens Interface-boks 534364.



Figur 25 Handling-stasjon



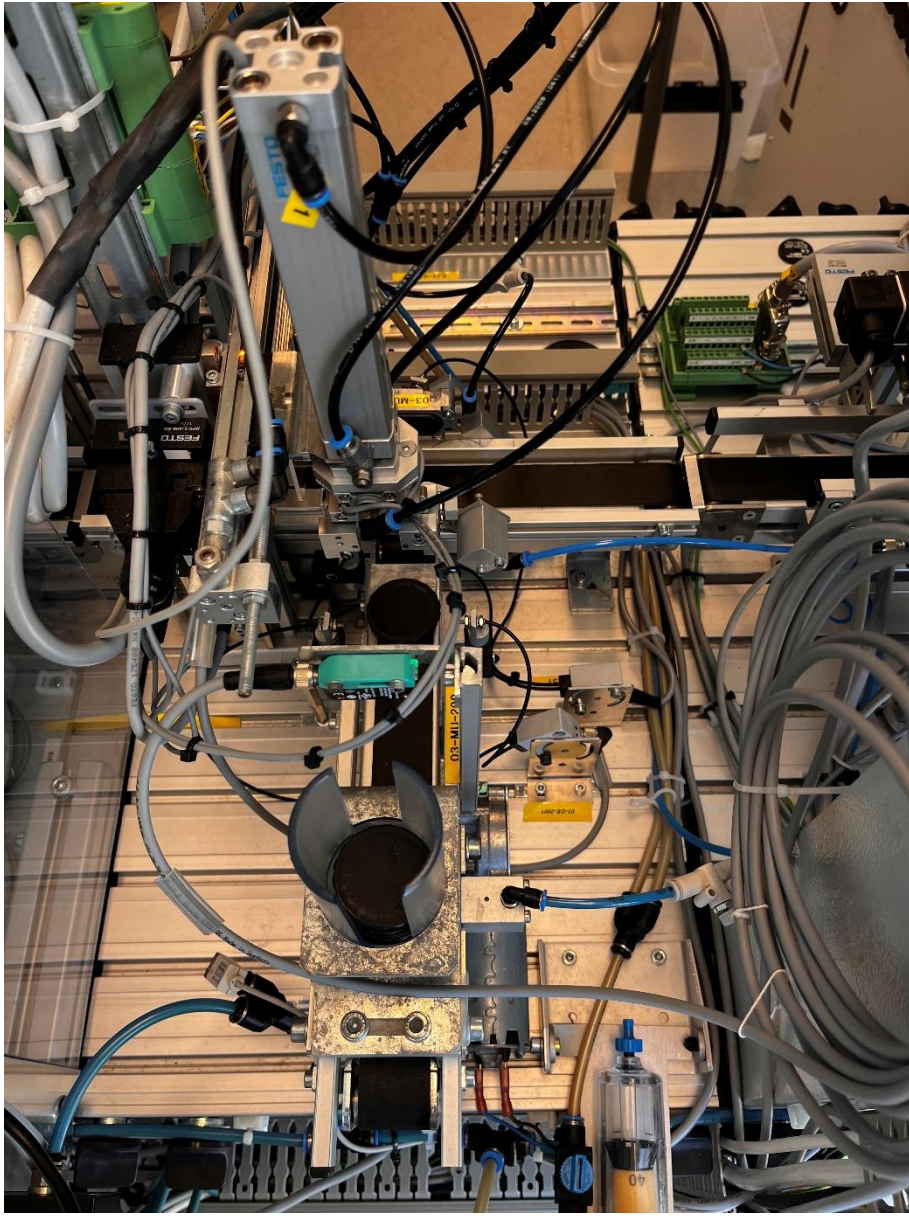
Figur 26 Varsellys.



Figur 27 I/O-moduler til fylling-PLS.



Figur 28 RFID-brikker.



Figur 29 Capping-stasjon.